

```
theory Week_1_sol
imports Main
begin
```

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Term 1

Computer-Assisted Theorem Proving: 7CCSACTP

Exercise Sheet 1

Important Note: Please download Isabelle from the link provided in the slides and bring a laptop to the large group tutorial. Performing the proofs in Isabelle will be an integral part of those tutorials.

Before you start, please open a new Isabelle/HOL theory file called Week_1.thy and add the following to its beginning.

```
theory Week_1
imports Main
begin
```

Exercise 1.1 Trying out Isabelle

In this sheet, you should try out the Isabelle theorem prover. Prove the following theorem, first on pen-and-paper, then formally in Isabelle.

```
lemma
  assumes T: T b
    and A: A a  $\wedge$  A b  $\implies$  a = b
    and TA:  $\bigwedge x. T x \implies A x$ 
    and Aab: A a
  shows T a
```

proof–

— *proof* — starts a structured proof without applying any method, leaving the goal unchanged.
— Style: declarative (Isar *have/show*) with apply-style steps; forward (*OF*); backward (*rule*); *subst*.

```
have 1: A b
  using TA[OF T]
```

```

.
— . proves the goal directly by a fact given with using, instantiating its variables if needed.

have 2:  $A\ a \wedge A\ b$ 
  apply(rule conjI)
    — rule r matches the conclusion of r with the goal and replaces the goal by r's premises
    (backward reasoning); here conjI leaves  $A\ a$  and  $A\ b$ .
  using  $Aab\ 1$ 
.

have 3:  $a = b$ 
  apply(rule A)
  using 2
.

show ?thesis
  using  $T$ 
  apply(subst 3)
    — subst eq rewrites the goal with the equation eq, left to right; here  $a$  becomes  $b$ .
.
qed

end

```